

1. Write a C program to simulate a Deterministic Finite Automata (DFA) for the given language representing strings that start with a and end with a

Code:

```
#include <stdio.h>
#include <string.h>

int main() {
    char s[100];
    printf("Enter string: ");
    scanf("%s", s);

    int n = strlen(s);

    if (n > 0 && s[0] == 'a' && s[n - 1] == 'a')
        printf("Accepted");
    else
        printf("Rejected");

    return 0;
}
```

Output:

```
1 // Online C compiler to run C program online
2 #include <stdio.h>
3 #include <string.h>
4
5 int main() {
6     char s[100];
7     printf("Enter string: ");
8     scanf("%s", s);
9
10    int n = strlen(s);
11
12    if (n > 0 && s[0] == 'a' && s[n - 1] == 'a')
13        printf("Accepted");
14    else
15        printf("Rejected");
16
17    return 0;
18 }
```

Enter string: abacda
Accepted

=== Code Execution Successful ===

